Residential asking-price prediction

NumPy decision trees, Pegasos and random Fourier features

AI Academy / National AI Center
MLE-AI-201 · Final Project V3

Selected classifier: F1 0.9218

Price model: median relative error 8.74%

Residential listings only · historical asking prices · held-out test n = 8,724

A declared residential scope and disjoint property groups

Subset	Rows	Property groups
Train	40,932	40,422
Validation	8,796	8,692
Test	8,724	8,622

Included

New / old apartments
Detached / garden houses

Excluded

Offices, commercial units, land and garages

Train-only encoding and scaling. Price-derived columns and owner/address identifiers never enter X.

Own trees search exact split thresholds

Core implementation

1. Sort candidate feature values
2. Score adjacent thresholds
3. Maximize impurity reduction
4. Recurse until stopping rules
5. Predict mean / majority at leaves

Selected price pipeline

Log-price target
Apartment / house routing
Train-only geographic distances
Depth 16 · minimum leaf 5
Exponentiate leaf predictions

Toy splits agree with sklearn; tests also cover zero-gain XOR splits and every stopping rule.

Own Pegasos learns linear and nonlinear margins

$$J(w, b) = \lambda\|w\|^2 / 2 + \text{mean}[\max(0, 1 - y(w \cdot x + b))]$$

Own model	λ	Epochs	Final objective	Margin 2/‖w‖
Linear	0.001	120	0.2380	0.559
RFF + linear	0.00001	120	0.2261	0.033

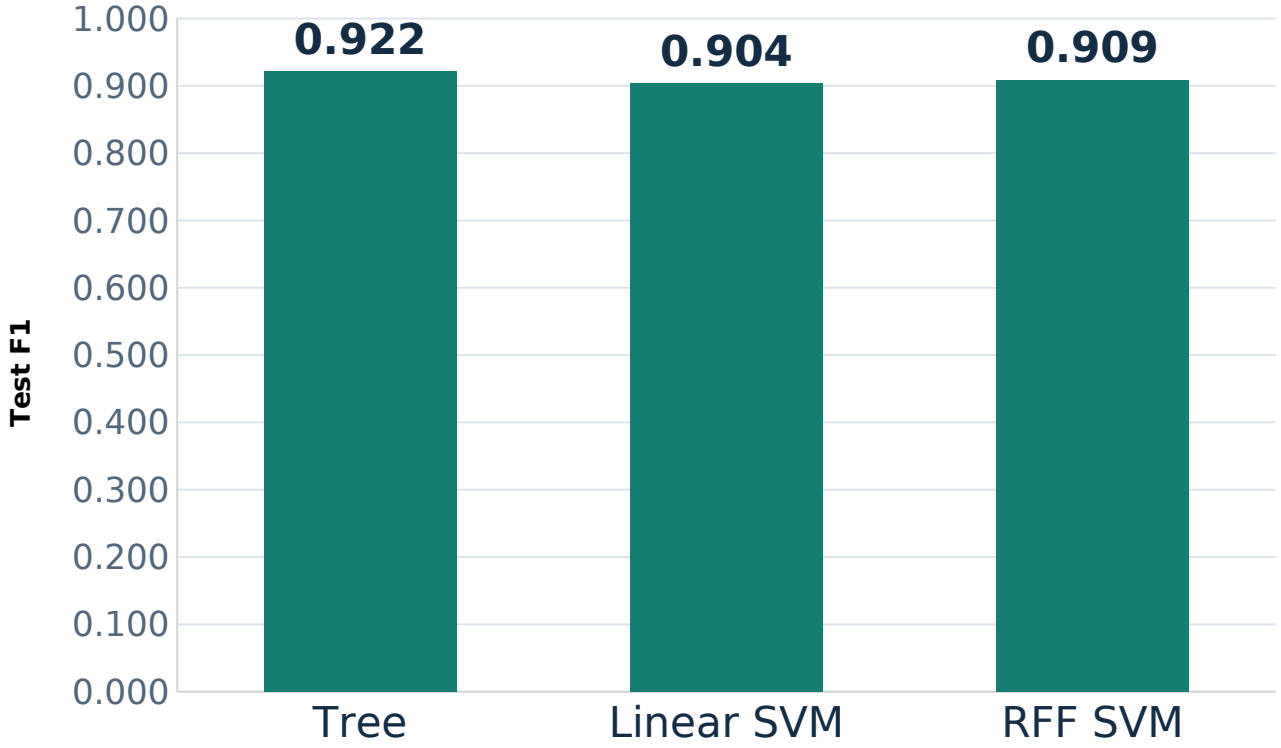
Nonlinear mapping: γ = 0.005, 2,048 Fourier features.
Zero-vector objective = 1.0; final objectives are lower.

Validation selects the configurations before test scoring

Candidate	Selected configuration	Validation
Price tree	Spatial + category routing; depth 16, leaf 5	169,875 AZN RMSE
Tier tree	Entropy; depth 16, leaf 10	0.9253 F1
Linear SVM	$\lambda = 0.001$; 120 epochs	0.9095 F1
RFF SVM	$\gamma = 0.005$; $\lambda = 0.00001$; 2,048 features	0.9136 F1

Frozen configuration → five-fold stability check → held-out test
Final training uses the original training subset only.

The selected tree reaches test F1 = 0.9218



Actual / predicted	Standard	Premium
Standard	3,981	308
Premium	380	4,055

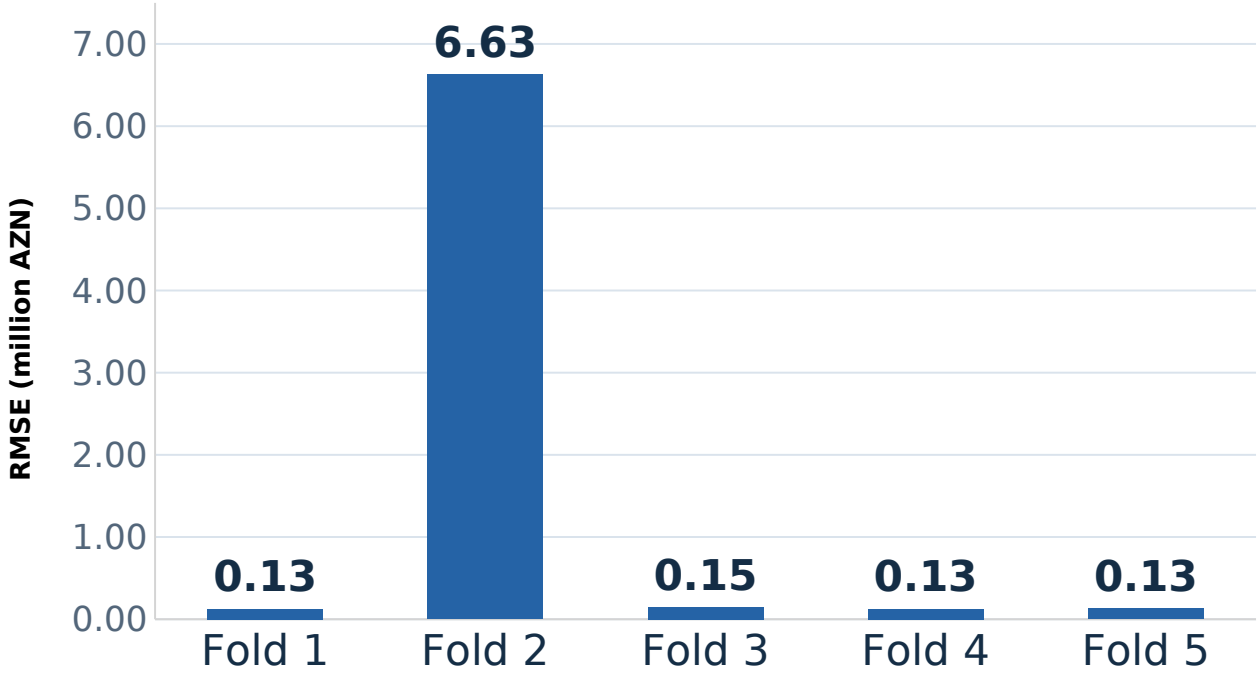
ROC AUC: 0.9670
Precision: 0.9294
Recall: 0.9143

Price accuracy is useful for typical rows, unstable in the tail

Model	RMSE (AZN)	MAE (AZN)	R^2
Train median	708,732	140,927	-0.010
Own spatial trees	667,507	56,568	0.104
sklearn spatial trees	666,545	56,592	0.106

Median absolute percentage error: 8.74%
77.4% of test prices are predicted within ±20%

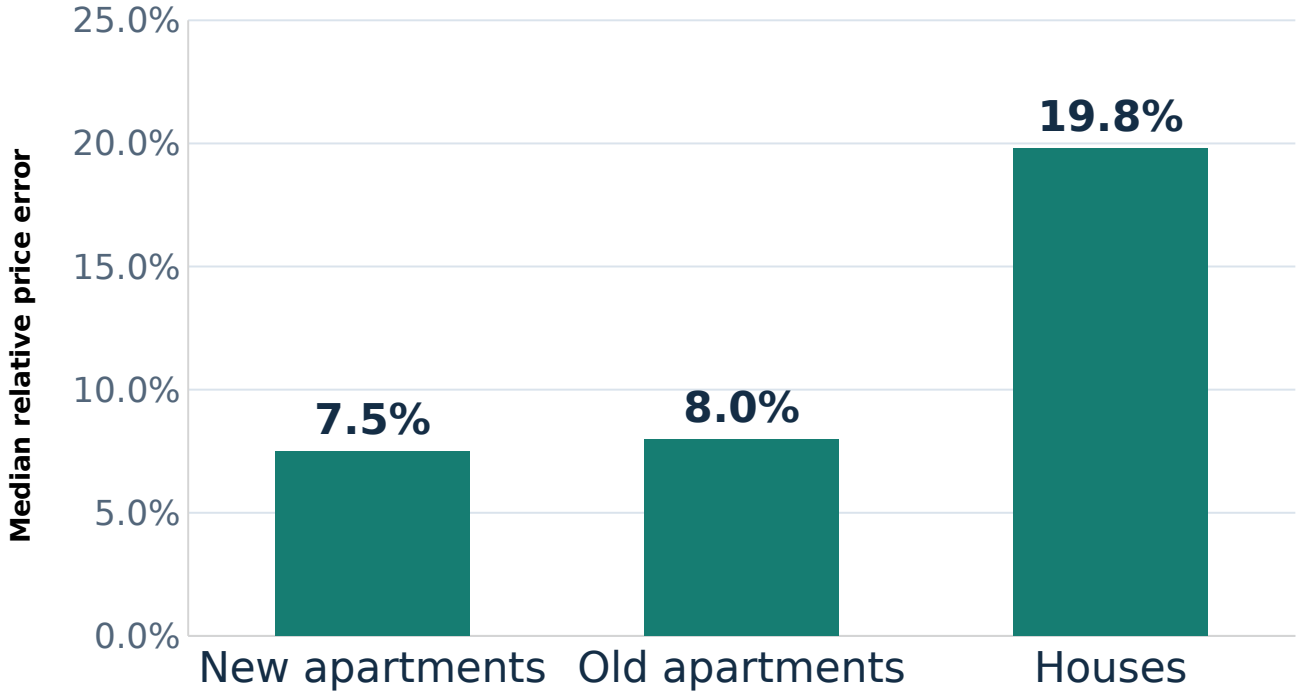
Grouped cross-validation exposes price-label risk



Classifier	CV F1 mean ± SD
Own tree	0.9183 ± 0.0023
Linear SVM	0.9061 ± 0.0038
RFF SVM	0.9111 ± 0.0023

Unverified extreme training price: 600,000,000 AZN.
Fixed configurations; CV is a stability check, not nested tuning.

Detached houses and threshold-adjacent prices are harder



Tier-classification error

28.0% near the threshold
5.2% farther away

Near = price within 10% of the 207,000 AZN threshold.

The worst 1% of test rows contribute 98.8% of squared price error. All rows stay in evaluation.

Matched sklearn baselines closely track the own models

Task / model	Own result	sklearn result
Price RMSE (AZN)	667,507	666,545
Tree F1	0.9218	0.9223
Linear SVM F1	0.9037	0.9032
RFF SVM F1	0.9088	0.9098

SVM scaling is inside Pipeline. Both RFF solvers share the same mapper.
Different optimizers and tree tie-breaking explain small gaps.

Reproducible code, honest limits and a short demo

python -m src.run_all

NumPy models + tests + saved estimators
Executed notebook + numerical outputs
IEEE-style report + editable slides

Historical asking prices; no independent transaction validation.
Future-period performance and extreme labels remain unresolved.

Developed with substantial ChatGPT/Codex assistance; human contributions must be confirmed by the team.